

EGGS



MANY KINDS OF ANIMAL, from earthworms and insects to fish and birds, reproduce by laying eggs. An egg is a single living cell complete with a supply of food. After the egg is laid, the cell starts to divide, and gradually a young animal's body takes shape. When the animal is ready to start life in the world outside, it breaks out of the egg, or hatches. There is a great variety of eggs – large and small, with shells and without. Some animals lay just a few eggs each time and look after them carefully. Others lay thousands or millions of eggs and leave them to develop on their own.



Eggs without a shell
Frogs' eggs do not have a shell. Instead, they are surrounded by a layer of jelly. The jelly swells up when the eggs are laid, forming a floating mass that can be more than 30 cm (12 in) across.



Eggs in strings
The common toad lays eggs like those of frogs, but they are laid in strings up to 3 m (10 ft) long. As the female lays the eggs, she winds them around underwater plants. The tadpoles hatch after about two weeks.



Leopard gecko's egg
Leathery eggs
Lizards and many other reptiles have eggs with a leathery shell. Unlike amphibians, reptiles can lay their eggs in dry places, because the shell helps keep the inside of the egg moist.



American robin's egg



Leatherhead's egg

Chalky eggs
The shell around birds' eggs is reinforced with a substance like chalk. To hatch, most young birds peck open their shell, but some kick their way out.



Cockroach egg package contains 16 eggs.

Packages

When cockroaches and praying mantises lay their eggs, they surround them with a special froth. The froth dries and hardens, forming a package like a tiny purse. This package protects the eggs, and the female carries it around on the end of her abdomen until the eggs are ready to hatch.

Mermaids' purses

Sharks lay some of the most unusual eggs. Instead of being round, their eggs can be flat, or even spiral. Dogfish, which are small sharks, lay eggs called "mermaids' purses". These have long tendrils with which the dogfish anchors the eggs to underwater plants.



Egg development

After an egg has been laid, a young animal starts to develop inside it. With some insects, such as the housefly, this can take less than a day, but with birds it may take more than a month. Eggs develop more quickly if they are warm, and most birds keep their eggs warm by sitting on them. This is called incubation.

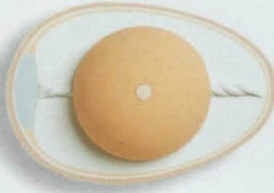


Mallee fowl

Instead of sitting on its eggs, the Australian mallee fowl buries them in a huge compost heap that it makes out of dead leaves. Heat from the giant heap keeps the eggs warm.

Development of a bird's egg

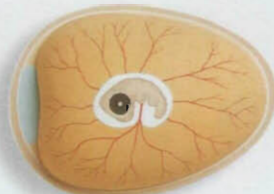
A bird's egg is divided into two main areas – the white and the yolk. The white is made of a substance called albumen. It stores water and cushions the developing chick from any sudden jolts. The yolk contains a store of food, which the chick uses up as it develops.



1 When the egg has just been laid, the part that will become the chick looks like a tiny pale spot. It lies on the upper surface of the yolk.



2 Within a day, cells in the spot start to divide to form an embryo. A network of blood vessels fans out over the yolk and supplies the embryo with food.



3 Three days after the egg was laid, the embryo is growing fast. Its eyes start to form, and tiny buds grow that will soon develop into wings and legs.



4 After seven days, the embryo has become a chick, and a special bag has formed to collect its waste. In three weeks, the chick's development will be complete.

Egg clutches

Some animals, such as queen termites, lay a steady stream of eggs, but most animals produce eggs in groups called clutches. The number of eggs in a clutch is closely linked to their size. For example, a wandering albatross has very big eggs, but it produces only one egg every two years. By contrast, a sunfish has tiny eggs, but it releases millions each time it breeds.

Giant clam
1,000,000,000



Wandering albatross
1



Cod
3,000,000



Mouth-brooding cichlid
6



Common octopus
150,000



Queen wasp
20,000



Common pheasant
8–15



Earwig
100

FIND OUT
MORE

ANIMAL
BEHAVIOUR

BIRDS

FISH

FROGS
AND TOADS

INSECTS

MAMMALS

NESTS AND
BURROWS

REPTILES

SHARKS
AND RAYS